Full Duplex UAV-Assisted Rate-Splitting Multiple Access Cellular Networks

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***Abstract*—This paper studies the downlink scenario of an unmanned aerial vehicle (UAV)-assisted rate-splitting multiple access (RSMA). The UAV serves as a full duplex (FD) amplify-and-forward relay to assist the base station (BS) in its communication with a set of user equipments (UEs). In this framework, we formulate an optimization problem with the goal of maximizing the minimum achievable rate by jointly optimizing the BS precoding vectors, the common-stream split, UAV transmit power, and the UAV location subject to the power budget constraints of the BS and UAV. Due to the non-convex nature of the problem, we propose an alternating optimization algorithm that decomposes the main problem into a power allocation subproblem and a UAV location subproblem, which are solved in an alternative way. Both subproblems are solved using a successive convex approximation approach. Our numerical results show that the proposed model outperforms traditional RSMA, non-orthogonal multiple access (NOMA), and UAV-assisted NOMA, demonstrating the efficacy of our approach in achieving higher minimum achievable rates.**

***Index Terms*—Amplify-and-forward, full-duplex, rate splitting multiple access, unmanned aerial vehicles.**

I. Introduction

The exponential growth in the number of connected devices and wireless traffic worldwide has created an unprecedented demand for next-generation wireless networks capable of accommodating such traffic [1]. This demand has been driven by the emergence of various highly demanding use cases and applications, such as augmented reality, human-to-brain interaction, and holographic telepresence [2]. Non-orthogonal multiple access (NOMA), a promising candidate for beyond 5G networks (B5G), has been extensively researched for its ability to increase spectral efficiency and support massive device connectivity compared to traditional orthogonal access techniques. By enabling multiple users to share the same resources simultaneously, NOMA holds great potential for future networks. However, selecting the decoding order of users, particularly in multiple antenna scenarios, remains a significant challenge in NOMA-based cellular networks [3].

Rate splitting multiple access (RSMA) is a promising access technology that incorporates some of the features of NOMA and has demonstrated greater resilience and generality, especially in multiple antenna situations [3]. RSMA functions by dividing the user's message into two components: a common part and a private part. The common part, which is chosen using a shared codebook, is distributed among all users, whereas the private part is specific to each user and independently decoded. The common stream is decoded initially, with each user handling all private streams as interference. Then, using the successive interference cancellation (SIC) technique, the common message is eliminated at each user's end, allowing the private message to be decoded while treating all other private streams as interference. RSMA is a versatile access technology that can be used in a variety of scenarios and has been shown to outperform other access technologies in certain situations [3], [4]. Specifically, the authors in [3] demonstrated the advantage of RSMA over the traditional NOMA and SDMA in terms of achievable sum rate. In addition, in terms of energy efficiency, RSMA proved to outperform the previously mentioned scheme [5]. The authors in [6], studied maximizing the sum rate in RSMA in the case of imperfect channel state information and showed its advantage.

Despite its benefits, in situations of severe channel attenuation, NOMA/RSMA may still suffer from degraded performance while serving users. To address this issue, cooperative communication techniques have been incorporated with NOMA/RSMA. Such techniques fall into two categories: 1) user-assisted: In this scenario, users with strong channels with the base station (BS) can assist others with poor channel conditions; 2) relay-assisted: In this scenario, a dedicated relay is deployed to forward the BS's signal to the users. Specifically, user-assisted RSMA is studied to enhance user fairness [7] and to minimize network energy consumption [8]. On the other hand, relay selection in cooperative NOMA is tackled in [9] to maximize network sum rate. Meanwhile, the authors in [10] study the use of a fixed relay to maximize the minimum user rate. It is worth mentioning that all aforementioned work, and citations therein, focused on having a fixed relay to assist the BS transmission. However, this strategy limits flexibility in dynamic environments, and thus, may degrade the overall performance. Starting from this problem, we consider a novel model that deploys a mobile relay that is capable of improving the RSMA networks. Specifically, our system leverages an unmanned aerial vehicle (UAV) as a relay due to its agility and ability to establish line of sight (LoS) communication, thereby enhancing performance in various applications [11].

Motivated by the above-mentioned gap, this paper formulates the problem of maximizing the minimum user rate by jointly optimizing beamforming vectors at the BS, and the UAV transmit power and location. Due to the high complexity of the problem and with the goal of solving the optimization problem in an efficient way, we invoke alternating optimization (AO) to divide the problem into two subproblems. The first one is to determine the precoding vectors at the BS and the trans-

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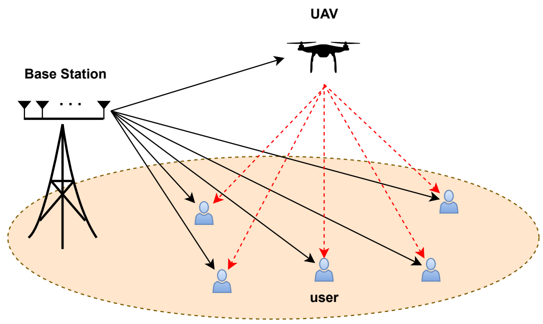


Fig. 1. System model.

mit power of the UAV. Meanwhile, the second subproblem tackles the UAV location. Finally, extensive simulation results demonstrated the advantage of our model over the traditional schemes.

II. System Model

A. Network Model

Our system model is depicted in Fig. 1 where we consider a downlink multiple-input single-output (MISO) system consisting of one BS equipped with N antennas, K numbers of users where each is equipped with a single antenna and a UAV that acts as an amplify and forward (AF) relay. We assume that the users may not have good channel conditions (e.g., weak or no presence of Line of Sight (LoS)) with the BS. Hence, to improve the signal quality on the users' sides, the UAV acts as a relay and amplifies the signal received from the BS in an FD mode. We define $\boldsymbol{h}_{b,k}$ as the channel coefficient between user-k and the BS. This channel is assumed to follow a path loss and Rayleigh fading such that $\boldsymbol{h}_{b,k} = \gamma_k^{-\frac{1}{2}} \boldsymbol{z}_k$, where $\gamma_k^{-\frac{1}{2}}$ and z_k denote the path loss and small scale fading [12], respectively. $\boldsymbol{h}_{b,u}$ represents the channel coefficient between the BS and the UAV, and $h_{u,k}$ denotes the channel coefficient between the user-k and the UAV. Both channels are assumed to follow a free space path loss as the UAV provides a clear LoS because of its altitude.

B. Transmission Model

The transmission model for the proposed system model consists of two phases that can be explained as follows:

Direct transmission: Utilizing the RSMA principle, the BS splits users' messages into common and private messages. All common messages from all users are combined together and form a combined common message. The combined message goes through an encoding process and creates a common stream. Meanwhile, the private messages of each user go through the encoding process independently and create private streams. Then, the BS superimposes the common and private streams and transmits the total signal to the users. This transmitted signal of the BS is received by the users and UAV.

Cooperative transmission: Utilizing AF protocol, UAV amplifies the signal received by the BS and sends it to the users. The users decode and combine the signal coming from the BS and UAV using the maximal ratio combining (MRC) technique [13]. Using SIC, each user removes the common stream from the total received signal and decodes its own private message.

Note that since the UAV operates in an FD relaying mode, both phases occur in the same time slot with the cost of self-interference (SI) at the UAV.

C. Signal Model

Based on the RSMA principle, the BS splits each user $k \in \mathcal{K}$ message, i.e., W_k, into common message $W_{c,k}$ and private message $W_{p,k}$, where $\mathcal{K}$ Is the set of users. Then, all the common messages are combined to form the common message denoted as W_c. Afterward, W_c is encoded into a common stream s_c; meanwhile, the private messages are encoded independently to create private streams $s_k, \forall k \in \mathcal{K}$. Accordingly, the overall data streams transmitted by the BS can be represented as $s = [s_c, s_k, \forall k \in K]^T$. These data streams are linearly precoded via the precoding vectors $\boldsymbol{P} = [\boldsymbol{p}_c, \boldsymbol{p}_k, \forall k \in K]$ where $\boldsymbol{p}_c$, $\boldsymbol{p}_k \in \mathbb{C}^{N\times 1}$. We can denote the final transmitted signal by $\boldsymbol{x}$ where $\boldsymbol{x} \in \mathbb{C}^{N\times 1}$. Thus, the transmitted signal by the BS can be expressed by,

$$\boldsymbol{x} = s_c \boldsymbol{p}_c + \sum_{k=1}^{K} s_p \boldsymbol{p}_k. \tag{1}$$

This signal is received by both the UAV and the users. Due to the FD relaying mode, the UAV can receive the signal from the BS and transmit the signal to the users simultaneously with the cost of a residual SI. That being said, the received signal at UAV can be expressed as,

$$y_u = \boldsymbol{h}_{b,u}^H \boldsymbol{x} + h_{SI}\sqrt{p_u}\hat{s}_c + n_u, \tag{2}$$

where p_u represents the transmit power of the UAV, $\hat{s}_c$ represents the delayed version of the signal that is relayed by the UAV to the users such that $\hat{s}_c = s_c(t-\tau)$, where t represents the t-th time slot and τ denotes the processing delay at the UAV [13]. It should be noted that the duration of processing time τ is negligible compared to the duration of one time slot. Also, this desynchronization allows the successful MRC on the user's side. h_{SI} is the SI channel at the UAV and it is assumed to follow a complex symmetric Gaussian random variable with a zero mean and variance Ω_{SI}^2. n_u represents the additive white Gaussian noise (AWGN) with zero mean and variance σ^2. The signal received by the direct transmission link is expressed as

$$y_{b,k} = \boldsymbol{h}_{b,k}\boldsymbol{x} + n_k, \tag{3}$$

while the signal received by the different users during the second transmission phase could be expressed as:

$$y_{u,k} = G\boldsymbol{h}_{b,u}^H \boldsymbol{h}_{b,k}\boldsymbol{x} + G\boldsymbol{h}_{b,k} n_u + G h_{si} + n_k, \tag{4}$$

where the gain, i.e., G, can be denoted as

$$G = \sqrt{\frac{P_u}{\sum_{k=1}^{K}|\boldsymbol{h}_{b,u}^H\boldsymbol{p}_k|^2 + |\boldsymbol{h}_{b,u}^H\boldsymbol{p}_c|^2 + p_u|h_{SI}|^2 + \sigma^2}}. \tag{5}$$

Here, n_k represents AWGN with zero mean and variance σ^2.

III. ACHIEVABLE RATE ANALYSIS

During the direct transmission, each user receives the common stream from the BS. Hence, the SINR at user k to decode the common stream coming from the BS can be written as

$$\Lambda^c_{k\to b} = \frac{|\boldsymbol{h}^H_{b,k}\boldsymbol{p}_c|^2}{\sum_{k'=1}^{K}|\boldsymbol{h}^H_{b,k}\boldsymbol{p}_{k'}|^2+1}. \tag{6}$$

Using AF technique, the UAV will receive the signal coming from the BS and amplify it. Thus, the SINR to decode it at the user k is denoted as:

$$\Lambda^c_{k\to u} = \frac{G^2|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}^H_{b,u}\boldsymbol{p}_c|^2}{G^2(\sum_{k'=1}^{K}|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}^H_{b,u}\boldsymbol{p}'_k|^2)+G^2|\boldsymbol{h}_{u,k}|^2 + G^2|\boldsymbol{h}_{u,k}|^2|h_{SI}|^2+1}, \tag{7}$$

which can be simplified to the expression

$$\Lambda^c_{k\to u} = \frac{p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}^H_{b,u}\boldsymbol{p}_c|^2}{D_1}, \tag{8}$$

where D_1 is defined as:

$$D_1 = \sum_{k'=1}^{K}(p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}^H_{b,u}\boldsymbol{p}_{k'}|^2) + p_u|\boldsymbol{h}_{u,k}|^2+|\boldsymbol{h}^H_{b,u}\boldsymbol{p}_c|^2+ \sum_{k'=1}^{K}|\boldsymbol{h}^H_{b,u}p_{k'}|^2 p_u|\boldsymbol{h}_{u,k}|^2|h_{SI}|^2+p_u|h_{SI}|^2+1. \tag{9}$$

To decode the common stream at the end of cooperative transmission, it is assumed that each user is fully capable of resolving the signal that is received from BS and UAV. Hence, the received signal at each user can be appropriately co-phased and merged by the MRC technique. Thus, the rate to decode the common stream is $R^c_{MRC,k} = \log_2(1+\Lambda^c_{k\to b}+\Lambda^c_{k\to u})$. However, this rate is subjected to the upper bound limit, i.e., R_T [14]. Therefore, the achievable rate to decode the common stream at each user can be given as $R^c_k = \min\{R_T, R^c_{MRC,k}\}$.

In order to ensure successful decoding of the common stream among all the users, the achievable data rate of the common stream should be constrained by the minimum of all the achievable common stream rates of all users, and it can be denoted as $R^c \le \min_{k\in K}\{R^c_k\}$. It should be noted that R^c is shared by all users. Hence, the portions of R^c are divided among all the users and can be expressed as follows

$$\sum_{k=1}^{K} c_k \le R^c, \tag{10}$$

where c_k is the portion of R_c that is allocated to user k. After decoding the common stream, each user removes it from the total received signal using the SIC technique. Hence, the SINR to decode the private message at each user from the direct transmission can be expressed as:

$$\Lambda^p_{k\to b} = \frac{|\boldsymbol{h}^H_{b,k}\boldsymbol{p}_k|^2}{\sum_{k'\neq k}^{K}|\boldsymbol{h}^H_{b,k}\boldsymbol{p}_{k'}|^2+1}. \tag{11}$$

Each user receives a message from the UAV. Hence, the SINR to decode this message could be expressed as:

$$\Lambda^p_{k\to u} = \frac{G^2|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}^H_{b,u}\boldsymbol{p}_k|^2}{G^2(\sum_{k'\neq k}^{K}|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}^H_{b,u}\boldsymbol{p}_{k'}|^2)+G^2|\boldsymbol{h}_{u,k}|^2 + G^2|\boldsymbol{h}_{u,k}|^2|h_{SI}|^2+1}, \tag{12}$$

which leads to the below expression:

$$\Lambda^p_{k\to u} = \frac{p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}^H_{b,u}\boldsymbol{p}_k|^2}{D_2}, \tag{13}$$

where D_2 can be defined as:

$$D_2 = \sum_{k'\neq k}^{K}\Big(p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}^H_{b,u}\boldsymbol{p}_{k'}|^2\Big) + p_u|\boldsymbol{h}_{u,k}|^2+|\boldsymbol{h}^H_{b,u}\boldsymbol{p}_c|^2+ \sum_{k'=1}^{K}|\boldsymbol{h}^H_{b,u}p_{k'}|^2 p_u|\boldsymbol{h}_{u,k}|^2|h_{SI}|^2+p_u|h_{SI}|^2+1. \tag{14}$$

By using the MRC technique, the two signals coming from the BS and the UAV are combined, yielding the rate to decode the private stream as $R^p_{MRC,k} = \log_2(1+\Lambda^p_{k\to b}+\Lambda^p_{k\to u})$. However, the achievable data rate to decode the private message has an upper bound R_T. So, it is rewritten as $R^p_k = \min\{R_T, R^p_{MRC,k}\}$. Thus, the achievable data rate for user-k can be obtained as.

$$R_{k,tot} = c_k + R^p_k. \tag{15}$$

IV. PROBLEM FORMULATION

In this subsection, the optimization problem is formulated with the objective of maximizing the minimum achievable rate by jointly optimizing the precoding matrix of the BS, the common stream portion, the UAV transmit power, and the UAV position, which can be expressed as:

$$\mathcal{P}_1: \max_{\boldsymbol{P},\boldsymbol{c},P_u,x_u,y_u} \min_{k\in\mathcal{K}}(c_k+R_{p,k}), \quad \forall k\in K \tag{16a}$$

$$\text{s.t.} \quad \sum_{k=1}^{K}\boldsymbol{p}^H_k\boldsymbol{p}_k+\boldsymbol{p}^H_c\boldsymbol{p}_c \le \mathcal{P}_T, \tag{16b}$$

$$0 \le P_u \le \mathcal{P}_{UAV}, \forall k\in K, \tag{16c}$$

$$c_k \ge 0, \forall k \in K, \tag{16d}$$

$$\sum_{k=1}^{K} c_k \le R^c, \tag{16e}$$

$$0 < x_u < x_{max}, \tag{16f}$$

$$0 < y_u < y_{max}, \tag{16g}$$

where $\boldsymbol{c} = [c_1, c_2, ...c_K]$ represents the common stream rate vector, and x_{max} and y_{max} represent the limit for the flying region of the UAV. (16b) and (16c) represent the power budget of BS and UAV, respectively. Constraints (16f) and (16g) are to ensure that the UAV is operating within a certain area. It is noticeable that $\mathcal{P}_1$ is a non-convex optimization problem due to the non-convex objective. Hence, it is necessary to divide problem $\mathcal{P}_1$ into some tractable subproblems that can be solved alternately and separately over multiple iterations. As a result, we divide the original problem into two subproblems. In the first subproblem, we jointly optimize precoding vectors, common stream split, and relaying power of the UAV. Meanwhile, in the second subproblem, we optimize the UAV placement. By invoking the AO algorithm, the two subproblems are solved in an alternative manner until convergence. The two subproblems will be discussed in detail in the next section.

V. Solution Approach

A. Solution approach for subproblem 1:

With given values of x_u and y_u, we study the joint optimization problem of precoding vectors, common stream split, and relaying power of UAV. At first, utilizing a slack variable α, we replace the objective function in $\mathcal{P}_1$ and reformulated it as follow:

$$\mathcal{P}_2 : \max_{\boldsymbol{P},\boldsymbol{c},P_u,\alpha} \alpha, \tag{17a}$$
$$\text{s.t.} \;\; (c_k + R_k^p) \geq \alpha, \;\; \forall k \in K, \tag{17b}$$
$$(16b) - (16e). \tag{17c}$$

However, non-convexity still exists in (16e) and in (17b). First, we deal with the non-convex constraints in (16e) by introducing two slack variables S_1 and S_2 such that

$$\sum_{k=1}^{K} c_k \leq \log_2(1 + S_1 + S_2), \tag{18}$$
$$\frac{|\boldsymbol{h}_{b,k}^H \boldsymbol{p}_c|^2}{\sum_{k'=1}^{K}|\boldsymbol{h}_{b,k}^H \boldsymbol{p}_{k'}|^2+1} \geq S_1, \tag{19}$$
$$\frac{p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}_{b,u}^H \boldsymbol{p}_c|^2}{D_1} \geq S_2. \tag{20}$$

It can be seen that (19) and (20) are still not convex. Starting with (19), we introduce another variable $\beta_{b,k}$. So, (19) could be reformulated as

$$\frac{|\boldsymbol{h}_{b,k}^H \boldsymbol{p}_c|^2}{\beta_{b,k}} \geq S_1, \tag{21}$$
$$\beta_{b,k} \geq \sum_{k'=1}^{K}|\boldsymbol{h}_{b,k}^H \boldsymbol{p}_{k'}|^2+1. \tag{22}$$

However, non-convexity still exists in (21). Therefore, we invoke first-order Taylor approximation to linearize it in an iterative manner. Particularly, the left side is approximated on two initial points $(\boldsymbol{p}_c^n, \beta_{b,k}^n)$, $\forall k \in K$. The approximated linear transformations can be given by [5],

$$\frac{2\mathcal{R}(\boldsymbol{p}_c^{nH} \boldsymbol{h}_{k,b}\boldsymbol{h}_{k,b}^H \boldsymbol{p}_c)}{\beta_{b,k}^n} - \frac{|\boldsymbol{h}_{k,b}^H \boldsymbol{p}_c^n|^2 \beta_{b,k}}{(\beta_{b,k}^n)^2} \geq S_1, \; \forall k \in \mathcal{K}. \tag{23}$$

Now, for (20), two slack variable S_3 and S_4 are introduced such that

$$S_3 \leq p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}_{b,u}^H \boldsymbol{p}_c|^2, \tag{24}$$
$$S_4 \leq D_1, \tag{25}$$
$$S_2 \leq \frac{S_3}{S_4}. \tag{26}$$

One can see that (24), (25) and (26) are still not convex. For (24), it is clear that there is multiplication of variables on the right-hand side of the equality [15]. Thus, we introduced the following constraints:

$$S_3 \leq |\boldsymbol{h}_{u,k}|^2 \phi_1^2, \tag{27}$$
$$\phi_1^2 \leq p_u|\boldsymbol{h}_{b,u}^H \boldsymbol{p}_c|^2. \tag{28}$$

Now, (28) is a quadratic conic convex constraint. On the other hand, (27) is still not convex. Hence, using first-order Taylor approximation, it could be approximated as follow:

$$f(\phi_1) = -(\phi_1^j)^2 - 2\phi_1^j(\phi_1 - \phi_1^j), \tag{29}$$

where ϕ_1^j is the value of ϕ_1 in the jth iteration. Now, for the constraint (25), we introduce another variable S_5 such that:

$$S_5 \leq \sum_{k'=1}^{K}\left(p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}_{b,u}^H \boldsymbol{p}_{k'}|^2\right), \tag{30}$$

$$S_4 \leq S_5 + p_u|\boldsymbol{h}_{u,k}|^2+|\boldsymbol{h}_{b,u}^H \boldsymbol{p}_c|^2+\sum_{k'=1}^{K}|\boldsymbol{h}_{b,u}^H p_{k'}|^2+ p_u|\boldsymbol{h}_{u,k}|^2|h_{SI}|^2+p_u|h_{SI}|^2+1. \tag{31}$$

Note that (30) can be handled in a similar way to (24). Then, for (26), it could be approximated by introducing two constraints as follow:

$$\eta \leq S_3, \tag{32}$$
$$(\frac{S_4}{a^n})^2 + (a^n S_2) \leq 2\eta, \tag{33}$$

where $a^n = \sqrt{\frac{S_4^n}{S_2^n}}$, and S_2^n and S_4^n are iterative variables. Meanwhile for (17b), it could be formulated by introducing the slack variables P_1 and P_2 as follow:

$$(c_k + P_1 + P_2) \geq \alpha, \tag{34}$$
$$\frac{|\boldsymbol{h}_{b,k}^H \boldsymbol{p}_k|^2}{\sum_{k'\neq k}^{K}|\boldsymbol{h}_{b,k}^H \boldsymbol{p}_{k'}|^2} \geq P_1, \tag{35}$$
$$\frac{p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}_{b,u}^H \boldsymbol{p}_k|^2}{D_2} \geq P_2. \tag{36}$$

One can see that (35) and (36) are non-convex. For that reason, We dealt with them in the same manner as (19) and (20), however, it is omitted for brevity.

B. Solution approach for subproblem 2:

In this subproblem, the UAV location subproblem is studied by fixing all the variables except x and y. Also, following the same approach of adding a slack variable to the objective function, the problem could be reformulated as:

$$\mathcal{P}_3 : \max_{x,y} \;\; \alpha, \tag{37a}$$
$$\text{s.t.} \;\; (c_k + R_k^p) \geq \alpha, \;\; \forall k \in K \tag{37b}$$
$$(16e) - (16g). \tag{37c}$$

In this subproblem, the non-convexity comes from the constraints (16e) and (37b). For that reason, a set of approximation techniques are applied to tackle this problem. Starting by (16e), it could be reformulated as follow:

$$\sum_{k=1}^{K} c_k \le \log_2\left(1+\frac{|\boldsymbol{h}_{b,k}^H\boldsymbol{p}_c|^2}{\sum_{k'=1}^{K}|\boldsymbol{h}_{b,k}^H\boldsymbol{p}_{k'}|^2+1}+S_7\right), \quad (38)$$

$$\frac{p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}_{b,u}^H\boldsymbol{p}_c|^2}{D_1} \ge S_7, \quad (39)$$

where S_7 is a slack variable. One can be seen that (39) is still non-convex. For that reason, we introduce two slack variables S_8 and S_9. So, (39) can be reformulated as:

$$\frac{S_8}{S_9} \ge S_7, \quad (40)$$

$$S_8 \le p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}_{b,u}^H\boldsymbol{p}_c|^2, \quad (41)$$

$$S_9 \le D_1. \quad (42)$$

The equations (40), (41) and (42) are still not convex with respect to x and y. The constraint (40) was tackled in the same way as (26). For (41), we introduce two slack variables S_{10} and S_{11} and the following constraints:

$$S_8 \le p_u S_{10} S_{11}, \quad (43)$$

$$\log_2(S_{10}) \le \log_2\left(\frac{\beta}{||L_k-L_u||^2+(H_k-H)^2}\right), \quad (44)$$

$$\log_2(S_{11}) \le \log_2\left(\frac{\beta}{||L_b-L_u||^2+(H_b-H)^2}|\mathbf{1}p_c|^2\right), \quad (45)$$

One can see that (43), (44) and (45) are still non-convex. Both (44) and (45) are tackled by introducing the slack vector $Y = [Y_k, Y_b]$ as follow

$$\log_2(S_{10}) \le \log_2\left(\frac{\beta}{Y_k+(H_b-H)^2}\right), \quad (46)$$

$$\log_2(S_{11}) \le \log_2\left(\frac{\beta}{Y_b+(H_b-H)^2}|\mathbf{1}p_c|^2\right), \quad (47)$$

where Y should satisfy the below condition:

$$Y \le ||L_i - L_u||^2, \quad (48)$$

where $i \in [k, b]$. And using first order Taylor approximation (with the iterative variable L_u^n), (48) is handled as follow:

$$Y_i \le ||L_i - L_u^n||^2 + 2(L_i - L_u^n)^T(L_u - L_u^n). \quad (49)$$

Now for (43), it could be solved by tackling the variable multiplication on the right-hand side of the equality. Thus, we introduced the constraint:

$$S_8 \le p_u\phi_2^2, \quad (50)$$

$$\phi_2^2 \le |\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}_{b,u}^H\boldsymbol{p}_c|^2. \quad (51)$$

Now, (51) is a quadratic conic convex constraint. On the other hand, (50) is still non-convex. Hence, by using first-order Taylor approximation over ϕ_2, it can be approximated as follow:

$$f(\phi_2) = -(\phi_2^j)^2 - 2\phi_2^j(\phi_2 - \phi_2^j), \quad (52)$$

Regarding (42), we introduce slack variables S_{12}, S_{13}, S_{14} and S_{15} such that:

$$S_9 \le S_{12} + S_{13}(p_u + p_u|h_{si}|^2) + S_{14} + S_{15} + p_u|h_{SI}|^2 + 1, \quad (53)$$

$$S_{12} \le \sum_{k=1}^{K}\left(p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}_{b,u}^H\boldsymbol{p}_k|^2\right), \quad (54)$$

$$S_{13} \le |\boldsymbol{h}_{u,k}|^2, \quad (55)$$

$$S_{14} \le |\boldsymbol{h}_{b,u}^H\boldsymbol{p}_c|^2, \quad (56)$$

$$S_{15} \le \sum_{k=1}^{K}|\boldsymbol{h}_{b,u}^H p_k|^2. \quad (57)$$

Note that (54), (55), (56), and (57) are non-convex. They are handled using the same way as (41) and (44). Finally, for (37b), it can be formulated by introducing slack variable P_3:

$$\left(c_k + \log_2\left(1+\frac{|\boldsymbol{h}_{b,k}^H\boldsymbol{p}_k|^2}{\sum_{k'\ne k}^{K}|\boldsymbol{h}_{b,k}^H\boldsymbol{p}_{k'}|^2}\right)+P_3\right) \ge \alpha, \quad (58)$$

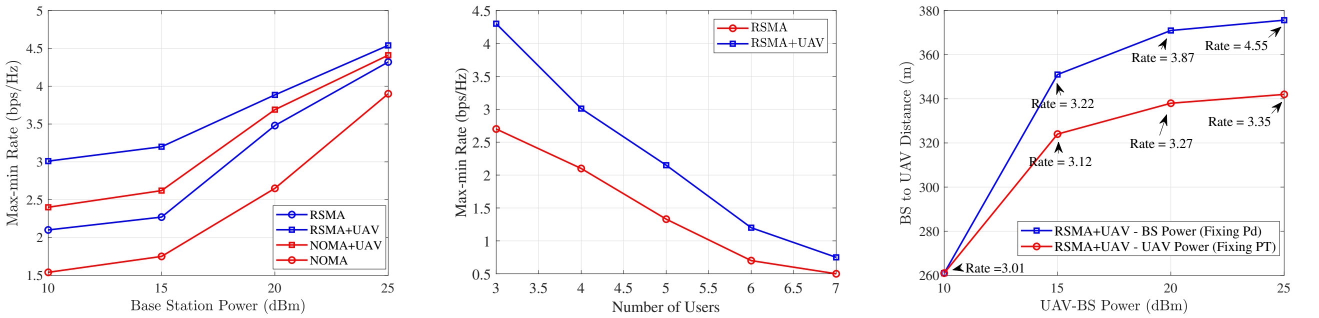
$$\frac{p_u|\boldsymbol{h}_{u,k}|^2|\boldsymbol{h}_{b,u}^H\boldsymbol{p}_k|^2}{D_2} \ge P_3. \quad (59)$$

One can see that (59) is still non-convex. For that reason, we dealt with them in the same manner as (39). The full details are eliminated for brevity. After having the two approximated subproblems, we invoke an AO algorithm that starts by fixing the UAV location and then obtaining the corresponding power variables, and vice versa until convergence.

VI. Numerical Evaluation

In this section, the numerical results are presented to illustrate and analyze the advantage of UAV-assisted RSMA. The performance is compared with the RSMA scheme without UAV, the NOMA scheme without UAV, and the UAV-assisted NOMA, where the UAV acts as an AF relay. For the UAV-assisted NOMA, we exhaustively enumerated the possible decoding orders in this regard and we select the optimal one. Unless stated otherwise, we consider the following parameters, the BS is equipped wit 4 antennae and has a transmit power $P_T = 10$ dBm, the UAV transmit power $P_u = 10$ dBm, $\sigma^2 = -174$ dBm/Hz, and the number of users is 4. Finally, our results are obtained by utilizing 150 independent Monte Carlo realizations.

Figure 2 (a) studies the max-min rate as a function of the BS power for the proposed model and the other baseline schemes. One can see that all the models present an increase in the max-min rate, with UAV-assisted RSMA showing the best performance. For instance, at 10 dBm transmit power, the UAV-assisted RSMA achieved about 30%, 42%, and 100% gain over UAV-assisted NOMA, RSMA, and NOMA, respectively. This is explained by the amplified signal of the UAV that enhances the signal received by the users and gives the BS more freedom in designing the precoding vectors. This gain then decreases with the increase of the transmit power of the BS. The reason behind this is the sufficient power provided



(a) Max-min rate vs the BS power.

(b) Max-min rate vs the number of users.

(c) BS and UAV distance vs the BS and UAV power.

Fig. 2. Numerical results for the model under different parameters.

by the BS, which leads to less dependence on the amplification exerted by the UAV.

In Figure 2 (b), we study the effect of the max-min rate as a function of the increase in the number of users. The graph demonstrates that the rate decrease with the increase in the number of users. This is because the resources of the BS are divided among several users, thus decreasing the power dedicated to each one. In addition, we can spot that the suggested model keeps a good gain compared to the RSMA model. For example, in the case of 7 users, the achieved gain is 25 percent. This is explained by the amplification of the UAV, which gives more flexibility to the BS in designing the beamformers, especially under an overloaded regime where the BS needs to serve many users.

Figure 2 (c) shows the distance between the BS and the UAV versus the BS power while fixing the UAV power and vice versa. It is clear that the UAV is getting away from the BS when we increase the BS's or UAV's power with enhanced rates. This is explained by the ability of the UAV to get closer to the users and hence provide more rates. On the other hand, we can see that for the same amount of power, the BS has a greater effect on the distance between the BS and UAV with a higher max-min rate. For instance, at 25 dBm, we can spot a significant difference between the two rates (4.55 and 3.35 bps/Hz). This can be interpreted by the fact that although the UAV has a higher transmit power, it loses the flexibility of moving and getting closer to the users to maintain a good link with the BS, and avoid harming itself with the interference.

VII. Conclusion

This paper presents a UAV-assisted RSMA model that operates in FD AF relaying mode. The model exploits the agility and ability of the UAV to establish an LoS communication to transmit the signal received by the BS. We aim at maximizing the minimum rate by jointly optimizing the BS precoding vectors, the common-stream split, and UAV transmit power and its location subject to the power budget constraints of the BS and UAV. Then, to tackle the non-convexity of the problem, an efficient alternating optimization algorithm is adopted by exploiting the SCA technique. The numerical results proved the advantage of using UAV as a relay in the RSMA network.

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